Fault-Ride Through of Cascaded Grid-forming (GFM) Inverters

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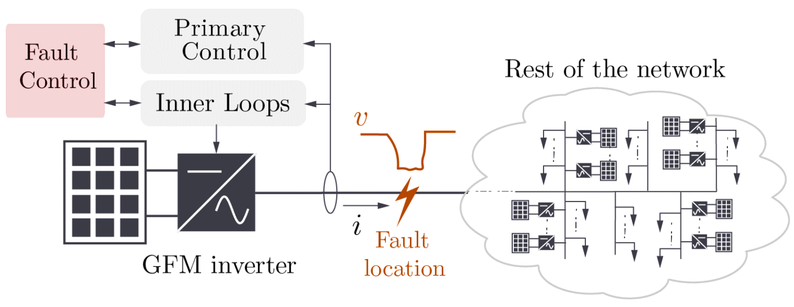
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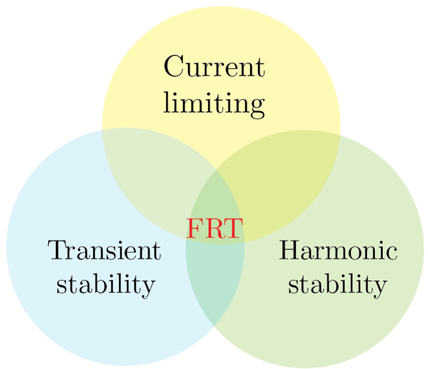
Motivation

- During a fault, GFM units may tend to inject high currents, potentially triggering a protective trip for the generation unit
- Cascaded converters has become popular for efficiently and reliably integrating low-voltage dc sources into the medium-voltage grid



What is FRT?

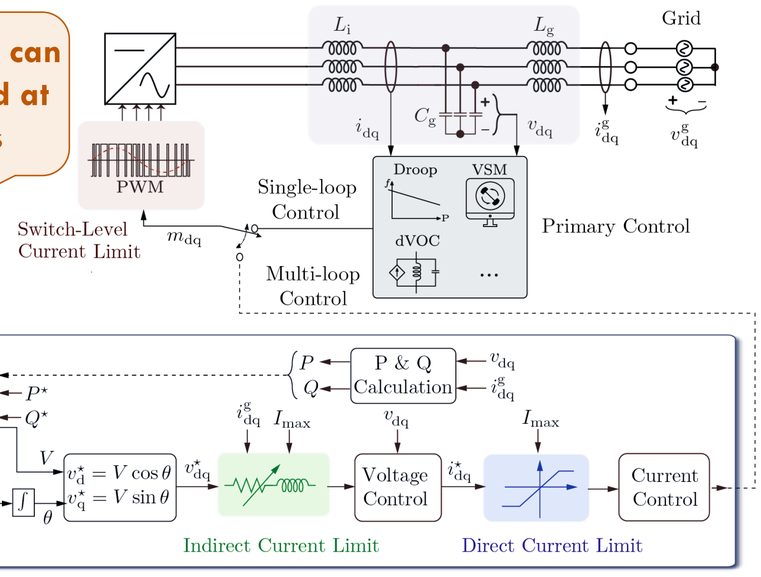
Current limiting ≠ FRT



GFM units should be able to stay in sync with grid during faults

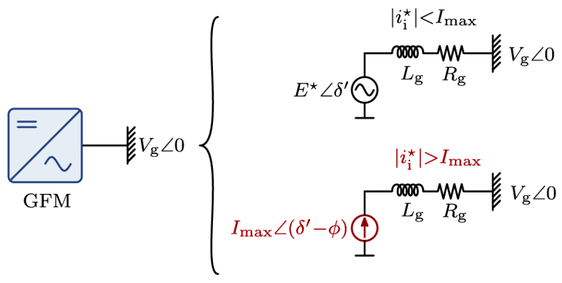
Overview of Current Limiters

Current limiters can be implemented at different stages

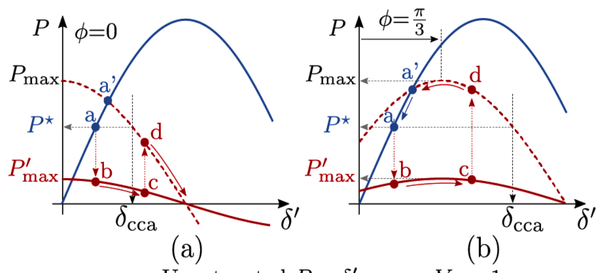


Direct Current Limiters

Current reference saturator



$$P=\begin{cases}\dfrac{EV_g}{X_g}\sin(\delta')=P_{max}\sin(\delta') & \text{if } |i_i^\star|<I_{max}\\ V_gI_{max}\cos(\delta'-\phi)=P'_{max}\cos(\delta'-\phi) & \text{if } |i_i^\star|>I_{max}\end{cases}$$

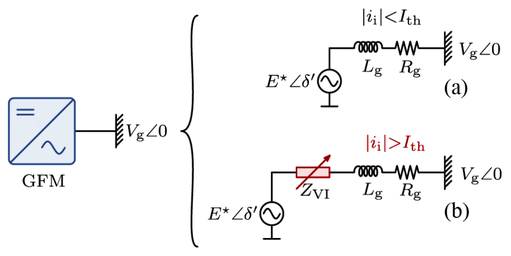


Unsaturated $P-\delta'$ curve, $V_g=1$ pu
Saturated $P-\delta'$ curve, $V_g=1$ pu
Saturated $P-\delta'$ curve, $V_g\ll 1$ pu

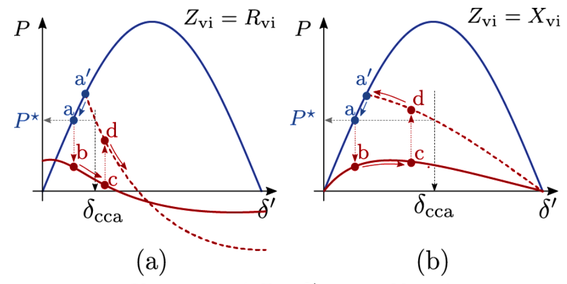
- Loses voltage-source nature, converts to a current source
- Poor Transient stability

Indirect Current Limiters

Virtual impedance-based limiter



$$P=\begin{cases}\dfrac{EV_g}{X_g}\sin(\delta')=P_{max}\sin(\delta') & \text{if } |i_i|<I_{th}\\ \dfrac{EV_g}{X_g+X_{vi}}\sin(\delta')=P'_{max}\sin(\delta') & \text{if } |i_i|>I_{th}\end{cases}$$

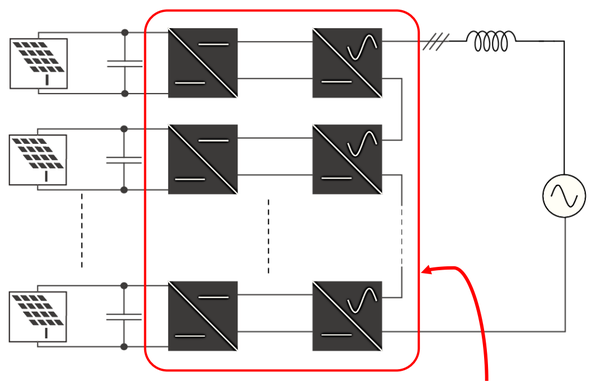


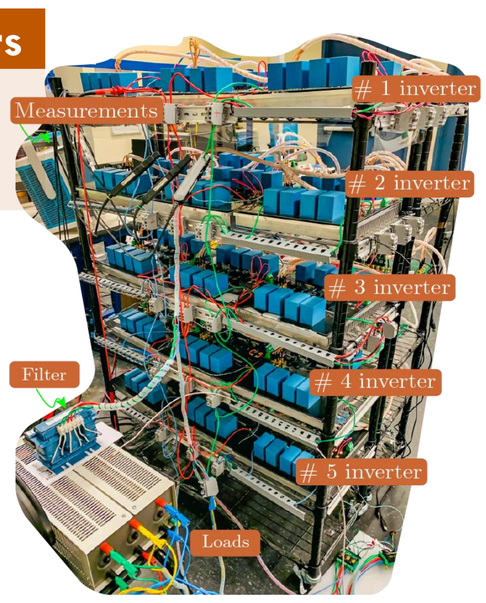
Unsaturated $P-\delta'$ curve, $V_g=1$ pu
Saturated $P-\delta'$ curve, $V_g=1$ pu
Saturated $P-\delta'$ curve, $V_g\ll 1$ pu

- Retains voltage-source nature
- Tuning R/X of virtual impedance is a challenge

Applications of Cascaded Inverters

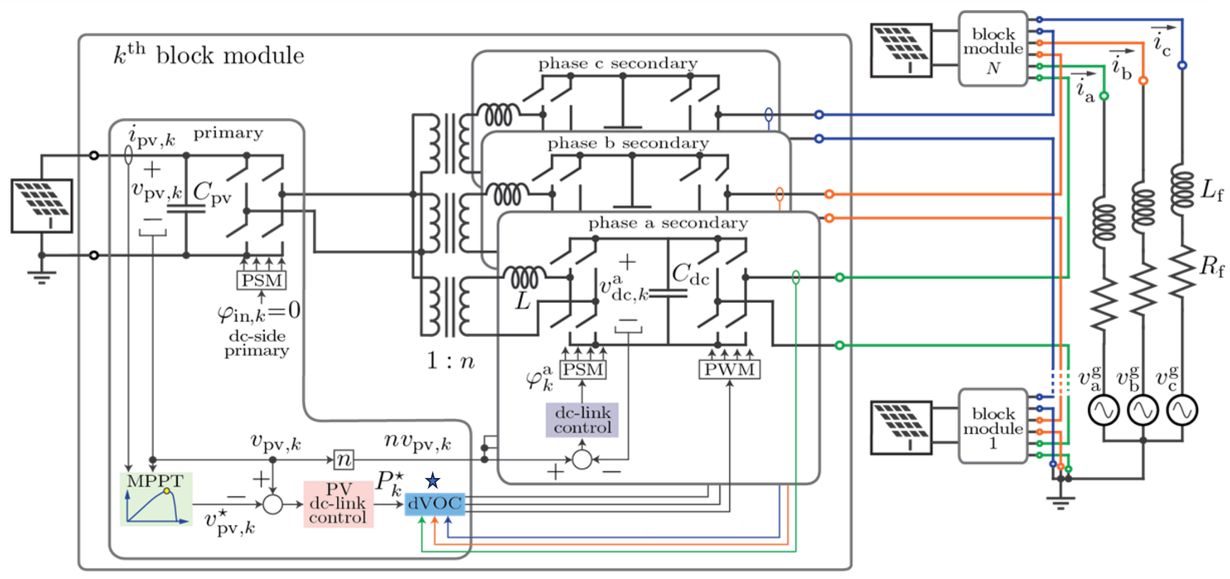
- Ultra-fast electric vehicle chargers
- Utility-scale solar and wind plants
- Solid-state transformers (SST)





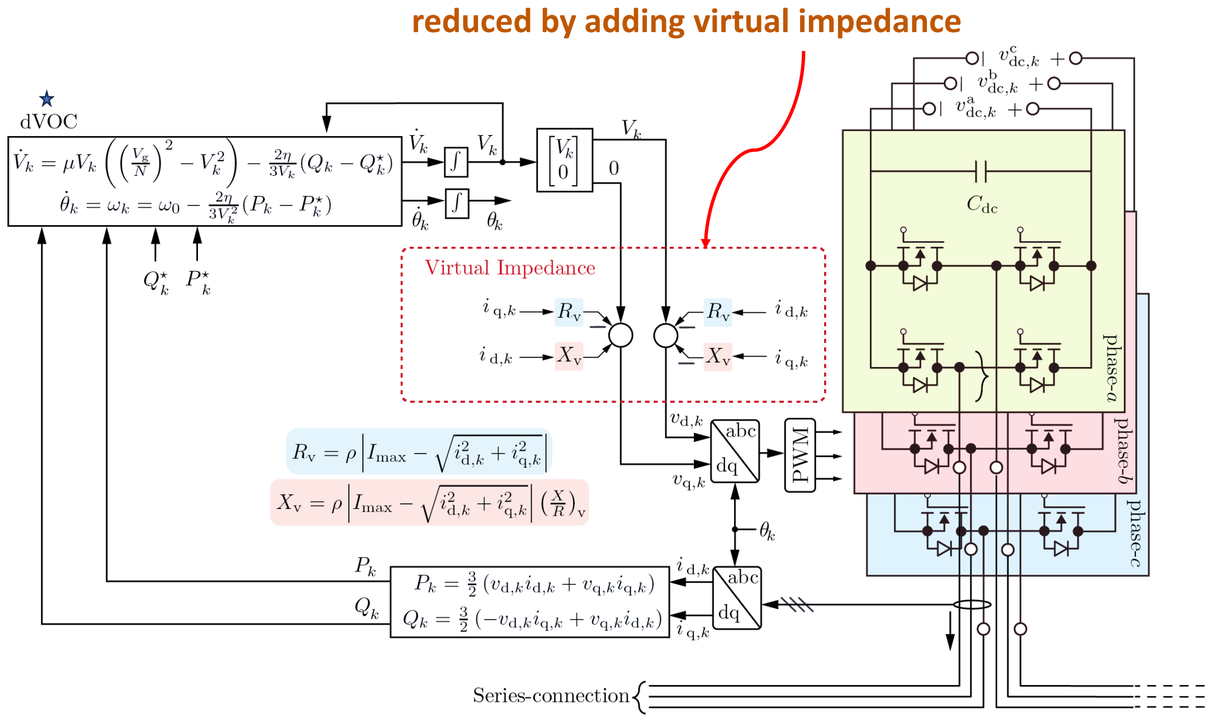
Avoids single point of failure, distributes voltage stress

Foundational Converter Block & its Controls

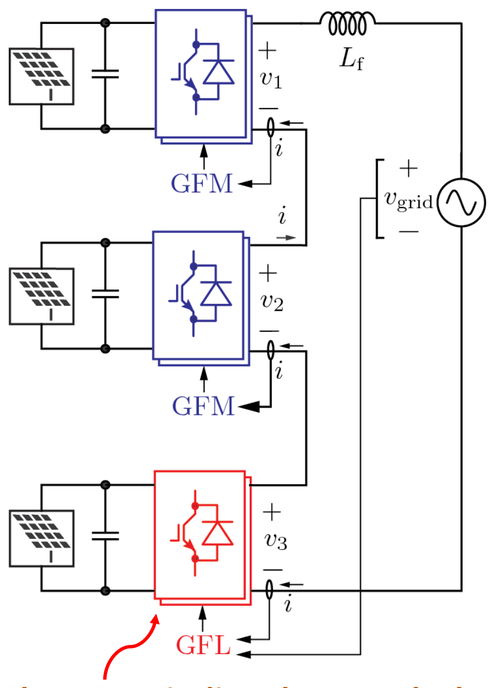


FRT Approach I-Virtual Impedance

★ **The voltage reference generated by the dVOC is reduced by adding virtual impedance**



Approach II-Mixed GFM-GFL



The GFL unit directly controls the current

Test Parameters

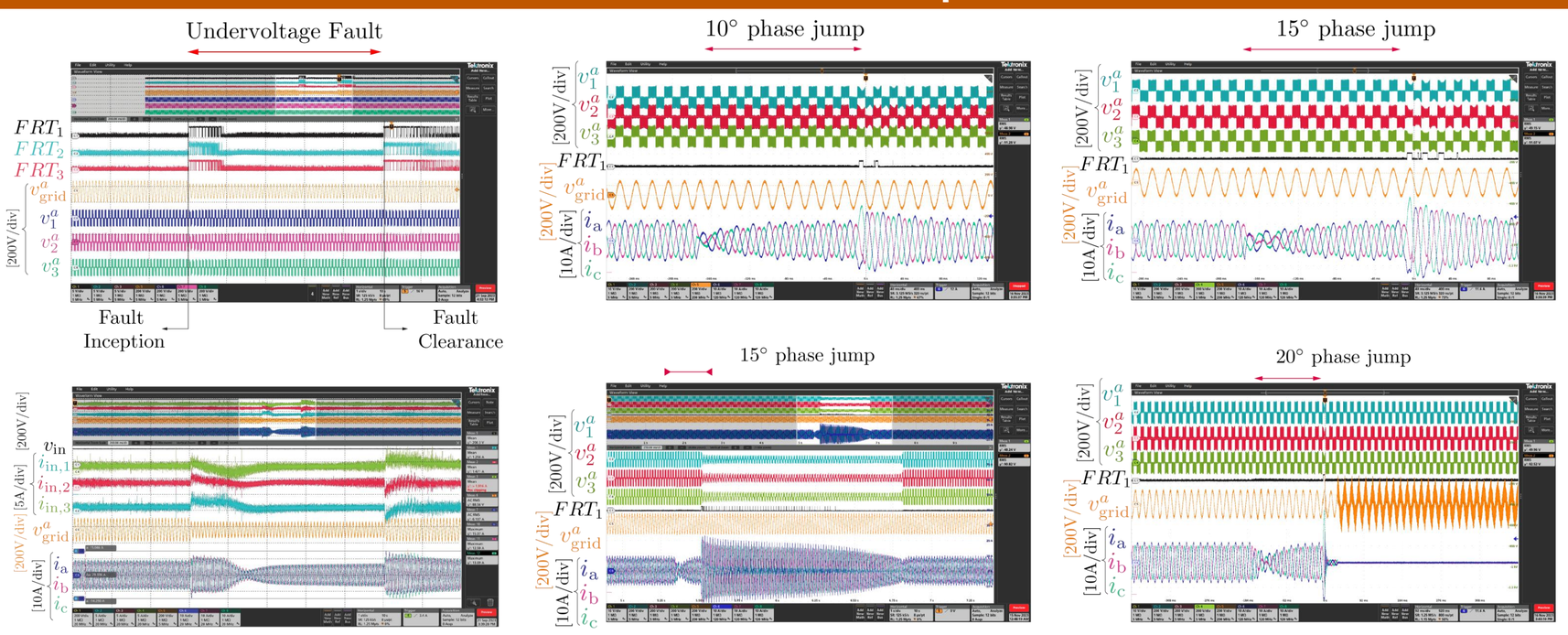
	Symbol	Description	Value	Units
Ac system	S_{rated}	module VA rating	1500	VA
	N	modules in series	3	
	V_g	rms line-neutral grid voltage	90	V
	ω_{nom}	rated frequency	60	Hz
	L_f	inductance	2.4	mH
	R_f	resistance	4.2	Ω
QAB dc-links	V_{dc}^0	nominal dc voltage	100	V
	ω_{sw}	switching frequency	100	kHz
	L	inductance	26	μH
	n	turns ratio	0.5	
	C_{pv}	input capacitance	660	μF
	C_{dc}	dc-link capacitance	200	μF
	$k_{p,dc}$	proportional gain	2.05	Ω^{-1}
	$k_{i,dc}$	integral gain	1290	$(\Omega s)^{-1}$

Conclusion and Future Work

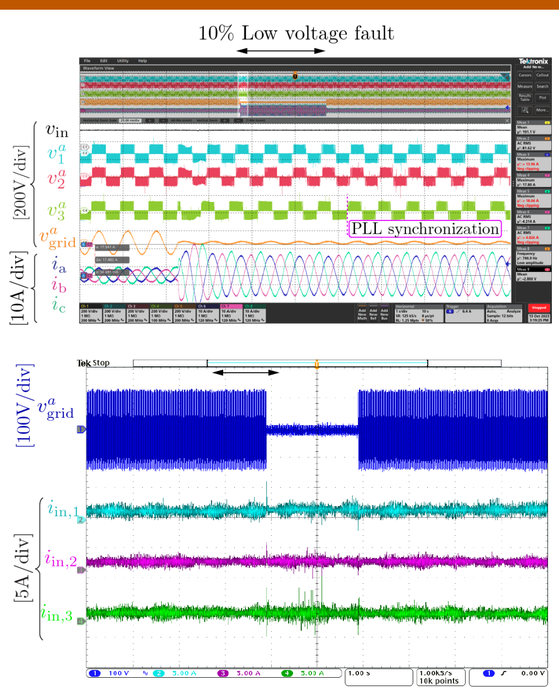
- Current limiter is designed for series-connected inverters
- Future work includes designing an unbalanced FRT strategy

Experimental Result

Approach I



Approach II



[1] R. Mallik, D. Chatterjee, and B. B. Johnson, "Fault Ride-Through of Series-Connected Grid-Forming Inverters," in preparation for IEEE TPEL

[2] N. Baeckeland★, D. Chatterjee★, M. Lu, B. Johnson, and G-S. Seo, "Overcurrent Limiting in Grid-Forming Inverters: A Comprehensive Review and Discussion," IEEE TPEL